

Algebra 1 Unit 14 Assessment Guide

General Information	Many of the benchmarks in the unit are supporting benchmarks. In other words, they are not explicitly taught but support the learning of the student. Much of the learning in the unit centers on MA.912.DP.3.1. Students are organizing categorical bivariate data into a frequency table. Students learn to consider possible associations which is further explored within the Honors benchmarks or in future courses. In the lessons, students learn how segmented bar graphs may be helpful in visualizing associations (MA.912.DP.1.1). Students interpret the data using the given information and organize the frequencies into two-way frequency tables. Teachers familiar with two-way tables may wish to extend a student's learning by having the student find relative frequencies. The benchmark does not require this and the draft test item specifications state that "Items will not require the student to find/interpret relative frequencies." The content of this unit, although new to the student, does not place a high cognitive demand on the student. Teachers may be tempted to skip teaching this content and replace it with reviews for the statewide end-of-course exam. Because most of the content is less demanding, teachers are encouraged to build student understanding to better equip students for success. Teachers may wish to consider that the information the teacher learned from the unit 13 assessment can be utilized to add in spiral learning to support students in areas of weakness while students have a less cognitively demanding unit.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1a	Benchmark(s)	MA.912.DP.3.1
3,175		Recommended Scoring (2 Total Points)	Consider no partial credit.
Question	1b	Benchmark(s)	MA.912.DP.3.1
3,287		Recommended Scoring (2 Total Points)	Consider no partial credit.
Question	1c	Benchmark(s)	MA.912.DP.3.1
1,411		Recommended Scoring (2 Total Points)	Consider no partial credit.
Question	1d	Benchmark(s)	MA.912.DP.3.1
3,248; 3,175; 3,136; and 3,287		Recommended Scoring (4 Total Points)	Consider no partial credit.

Question	2a			Benchmark(s)	MA.912.DP.3.1, MA.912.DP.1.2																
<table><tr><td></td><th>Flycatcher</th><th>Hummingbird</th><th>Total</th></tr><tr><th>Cacao Farm</th><td>64</td><td>38</td><td>102</td></tr><tr><th>Rainforest</th><td>19</td><td>37</td><td>56</td></tr><tr><th>Total</th><td>83</td><td>75</td><td>158</td></tr></table>					Flycatcher	Hummingbird	Total	Cacao Farm	64	38	102	Rainforest	19	37	56	Total	83	75	158	Recommended Scoring (18 Total Points)	Consider giving students 2 points for each value in the table.
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Rainforest	19	37	56																		
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Question	2b			Benchmark(s)	MA.912.DP.3.1, MA.912.DP.1.2																
There is an association between the bird’s location and the abundance of birds. The abundance of hummingbirds in the rainforest and on the cacao farm is approximately equal, while the abundance of flycatchers on the cacao farm is almost $3\frac{1}{2}$ times that of the abundance of flycatchers in the rainforest. The abundance of hummingbird’s is not affected by location, but the abundance of flycatchers is.				Recommended Scoring (3 Total Points)	Consider no partial credit.																

Question		3a				Benchmark(s)	MA.912.DP.3.1																																																																																	
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Question	4a			Benchmark(s)	MA.912.DP.3.1	
		Has a Job	Does Not Have a Job	Total	Recommended Scoring (10 Total Points)	Consider giving students 2 points for each missing value in the table.
Has Driver's License	1,350	450	1,800			
Does Not Have a Driver's License	40	160	200			
Total	1390	610	2,000			
Question	4b			Benchmark(s)	MA.912.DP.3.1, MA.912.DP.1.2, MA.912.DP.1.3	
About $\frac{3}{4}$ or the seniors surveyed that have a driver's license have a job, whereas only about $\frac{1}{5}$ of those who don't have a license have a job. This shows that there is an association between having a driver's license and having a job.				Recommended Scoring (3 Total Points)	Consider no partial credit.	